

Weekly Report: [W09]

General Information

- **Group ID:** 52
- **Group Name:** Group 52
- **Class:** 25A01
- **Project Name:** Super Mario Game
- **Date range:** 2026-08-02 – 2026-08-08
- **Github Repository:** <https://github.com/ndmhuy/SuperMarioGame.git>

Tasks Completed This Week

25125083 - Nguyễn Đình Minh Huy (Member A - Engine, Physics & Player/Item Entities)

- **Completed Tasks:** No committed changes this week.

Note: He did not push the commit to the remote.

25125084 - Trần Gia Huy (Member B - Input, Sound & Gameplay Systems)

Branch: `B/feature/graphic-visual` (Graphics, Spritesheets & Visual FX)

- **Task 1: Complete Spritesheets Integration:** Standardized and imported complete sprite sheet coordinates and graphics for enemies, items, and scenery elements. Configured `enemy_projectile.json` (153 frames) and `item.json` (49 frames). Authored `SPRITE_NAMING_CONVENTIONS.md` detailing structural standards.
- **Task 2: Interactive Spritesheet Studio & Merger Tools:** Developed web utility interfaces (`spritesheet_merger.html` , `spritesheet_studio.html` , `clean_outlines.html`) and helper Python scripts (`merge_spritesheets.py` , `clean_spritesheet_outlines.py`) to automate texture grid merging, bounds verification, and outline cleaning.

- **Task 3: Interactive Enemy Visual Tester:** Coded verify_enemies_visual.cpp establishing a standalone GUI environment to play and test enemy animations (Goomba, Koopa, Boo, etc.) with real-time ImGui configurations.
- **Task 4: Complete Transformation & Color Filter Animations (Tasks 5.8 & 5.9):**
 - Created SpriteColorFilter supporting HSL-to-RGB rainbow power-up cycling, damage alpha modulation, and initial hit flash tinting.
 - Created SpriteTransformAnim supporting easing scale transformations (Mega/Mini size transitions) anchored at sprite feet bottom-center, spin rotation, and transformation form flickering.
 - Created EntityDeathEffect managing launch velocity and rotational physics for stomps, shell launches, and player freeze-hop falls.
 - Created ScreenTransitionManager supporting circular viewport scale fade in/out transitions and solid color flashes.

Branch: `B/feature/sfx-music` **(Audio and Sound Systems)**

- **Task 1: Sound Manager & Fallback Mechanism:** Programmed SoundManager with a fallback buffer to dynamically generate beep waveforms if sound files are missing, protecting against runtime crashes.
- **Task 2: Import Gameplay Sound Effects:** Added 29 WAV audio files under `assets/sfx/` representing player jumps, block breaks, powerups, flagpole, stamp, lava fall, and pipe warp. Authored REQUIRED_SFX.md mapping spec requirements.
- **Task 3: Interactive Sound Visual Tester:** Programmed verify_sound_visual.cpp supplying a test GUI displaying play state controls.

AI Usage Declaration

- Used AI to design HSL-to-RGB rainbow cycling math and alpha flickering intervals inside `SpriteColorFilter`.
- Used AI to structure bottom-center layout transform origin anchoring inside `SpriteTransformAnim`.
- Used AI to design a procedural sine-wave synthesizer to produce fallback beep buffers in `SoundManager`.

- Used AI to program the Canvas-based pixel border extraction in `clean_outlines.html`.
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Tasks Planned for Next Week

- **Nguyễn Đình Minh Huy:** Continue Phase 7 (Game States, Main Menu UI, World Map selection layers) and integrate Time Rewind circular buffer.
 - **Trần Gia Huy:** Complete Task 5.11 (Wire Sprites and Animations to Main Game System) by staging and committing the local changes, verifying all entity renders, and testing tile animations. Begin Phase 10 console debug overlays.
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Issues & Resolutions

Issue 1: Out-of-Bounds Sprite Artifacts & Border Outlines

- **Problem:** When importing custom sprite frames from different texture packages, color outlines and borders around sprite frames had visual artifacts (e.g., non-transparent outlines, mismatched scaling, or alignment issues).
- **Resolution:** Developed `scripts/merge_spritesheets.py` and `tools/outline-cleaner/clean_outlines.html` to programmatically strip dark borders, filter alpha values, and stitch individual grid cells into optimized sheets (`item.png` and `enemy_projectile.png`).

Issue 2: Audio Asset Failure Safety

- **Problem:** In SFML, attempting to load a missing or corrupted audio file using `sf::SoundBuffer::loadFromFile()` would fail silently or throw exceptions at runtime, causing a crash when play is requested.
- **Resolution:** Implemented a fallback mechanism in `SoundManager::loadSound()`. If an audio asset is missing, it dynamically synthesizes a basic 10ms beep waveform programmatically into an in-memory buffer (`m_fallbackBuffer`), ensuring safety against missing file crashes.

Side note: If there is a newer version of the report, it will be the valid one and this will be outdated.